What is GEMM?

Definition: Single-precision general matrix multiplication (SGEMM)

$$C \leftarrow \alpha AB + \beta C$$

- Setting $\alpha = 1$ and $\beta = 0$ gives ordinary matrix multiplication: $C = AB$.
- Other values let one call scale the product and accumulate it into an existing matrix.

Example

During the forward pass when training a model, we accumulate the gradients from the entire batch.

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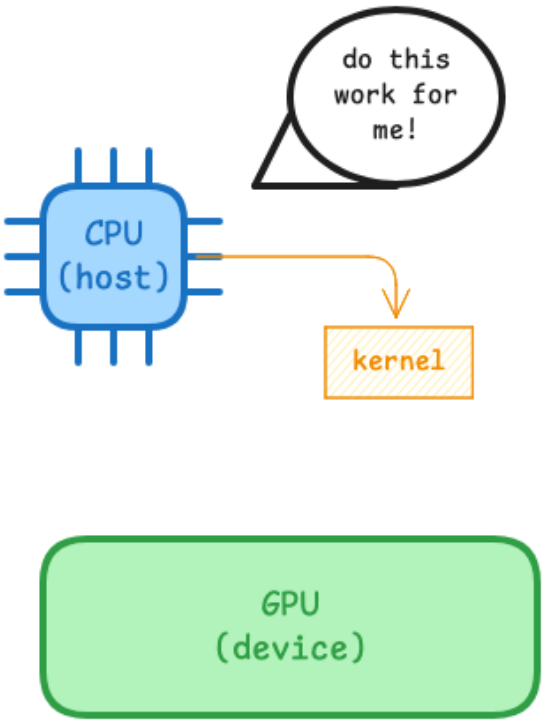
```
sgemm(
  int M, int N, int K,
  float alpha, float *A, float *B,
  float beta, float *C
)
```

$$\underbrace{A}_{M \times K} \quad \underbrace{B}_{K \times N} \quad \underbrace{C}_{M \times N}$$

- M: rows of A and C
- N: columns of B and C
- K: shared reduction dimension

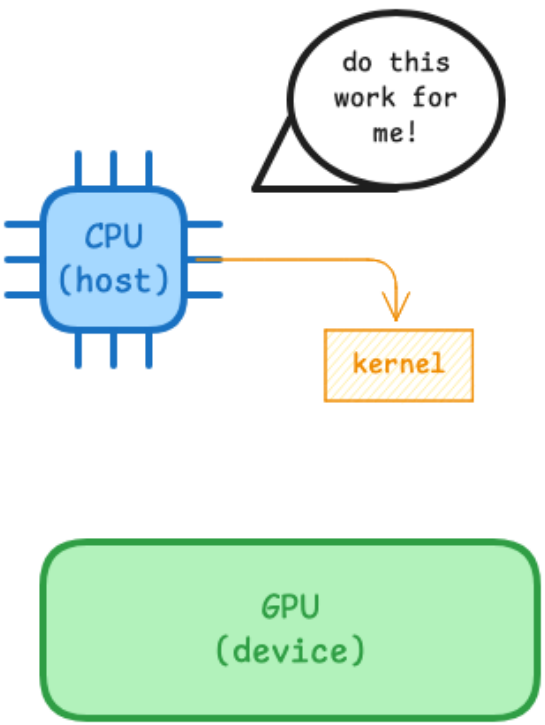
CUDA Basics (Host vs. Device)

- The CPU is known as the **host**, and manages top level logic.
- The GPU is known as the **device**, and does the heavy lifting.
- The host launches **kernels** to the device.



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- The CPU is known as the **host**, and manages top level logic.
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```
int main() {
  const int M = 128, N = 128, K = 128;

  float *dA, *dB, *dC;
  cudaMalloc(&dA, M * K * sizeof(float);
  cudaMalloc(&dB, K * N * sizeof(float));
  cudaMalloc(&dC, M * N * sizeof(float));

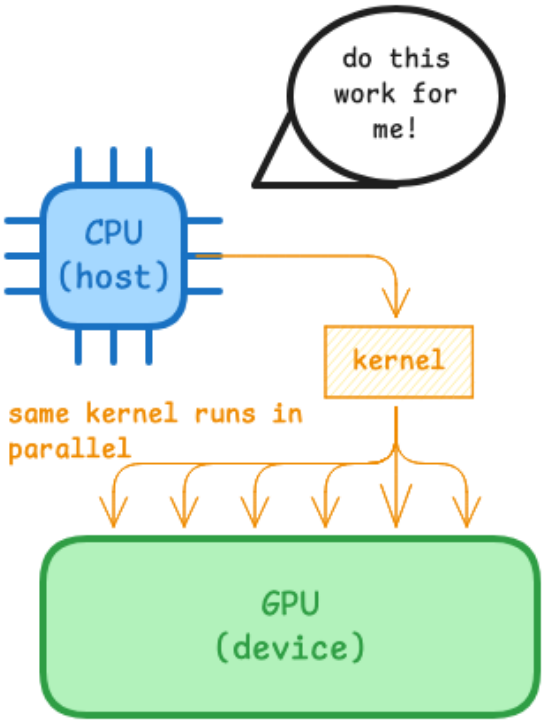
  // load values into these device pointers...

  run_sgemm_naive(M, N, K, 1, dA, dB, 0, dC);

  cudaFree(dA); cudaFree(dB); cudaFree(dC);
}
```

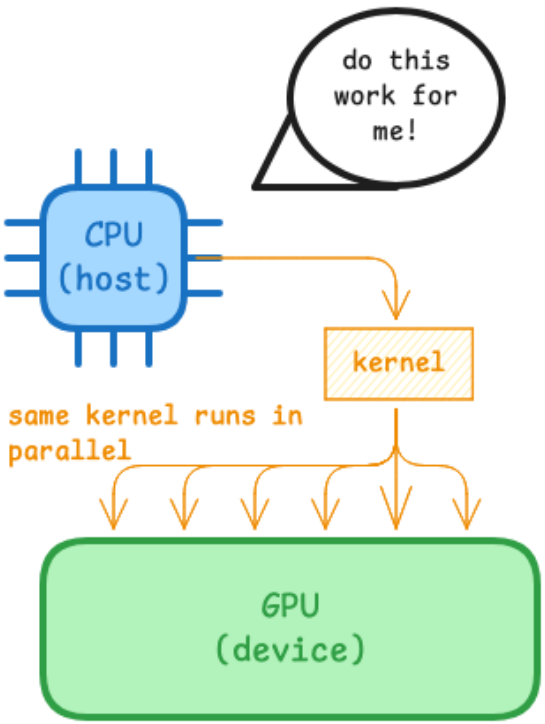
CUDA Basics (Kernel Execution)

- A kernel runs in parallel on many **threads** on the device.
- The number of threads the device allocates is configured by the host, when it **launches** the kernel.



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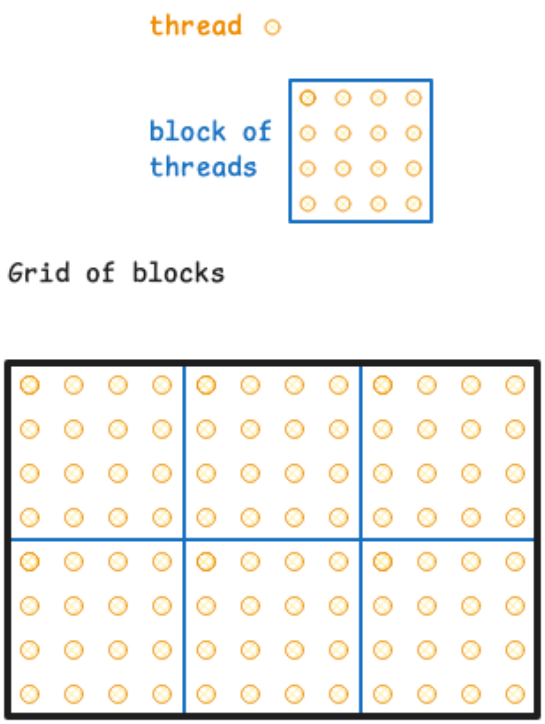



```
__global__ void sgemm_naive(
  int M, int N, int K, float alpha, const float *A,
  const float *B, float beta, float *C
) {
  // implement kernel ...
}

void run_sgemm_naive(int M, int N, int K, float alpha,
     float *A, float *B,
                        float beta, float *C) {
  dim3 blockDim(32, 32);
  dim3 gridDim(CEIL_DIV(M, 32), CEIL_DIV(N, 32));
  sgemm_naive<<<gridDim, blockDim>>>(
      M, N, K, alpha, A, B, beta, C);
}
```

CUDA Basics (Threads, Blocks, Grids)

- Threads are grouped into **blocks**.
- The collection of blocks requested by the host is organized in a **grid**.




```
__global__ void sgemm_naive(
  int M, int N, int K, float alpha, const float *A,
  const float *B, float beta, float *C
) {
  // implement kernel ...
}
```

```
void run_sgemm_naive(int M, int N, int K, float alpha,
     float *A, float *B,
                       float beta, float *C) {
  dim3 blockDim(32, 32);
  dim3 gridDim(CEIL_DIV(M, 32), CEIL_DIV(N, 32));
  sgemm_naive<<<gridDim, blockDim>>>(
      M, N, K, alpha, A, B, beta, C);
}
```

Why Threads, Blocks, Grids?

This seems complicated, why can't we just ask for N threads?

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Relation to hardware

This hierarchy is closely related to how the device schedules the execution. We will see later how to use this to optimize performance.

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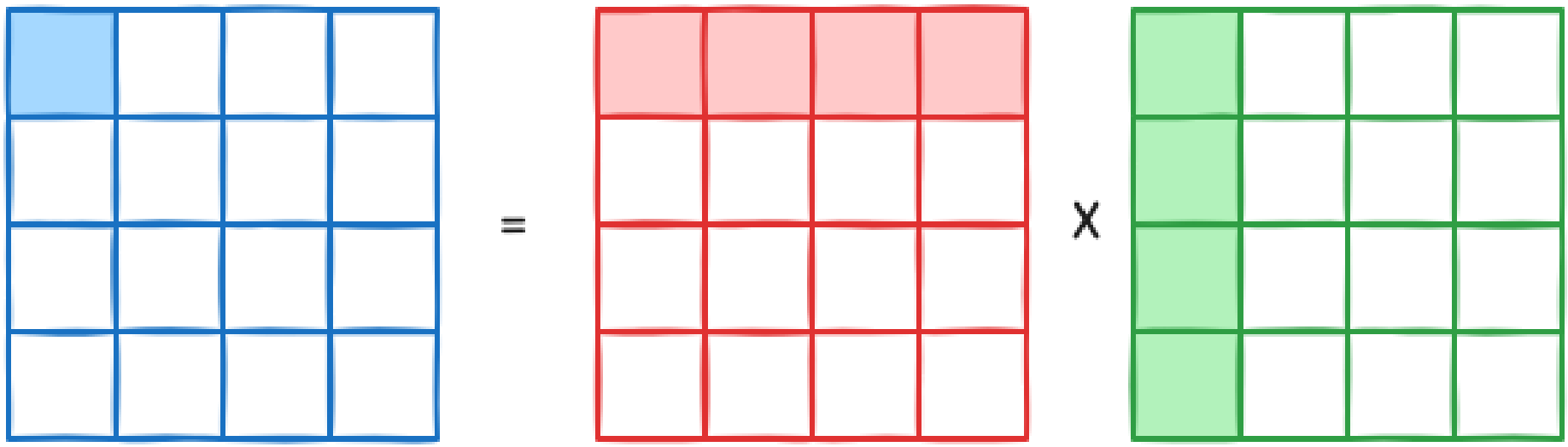
Thread, Block, Grid hierarchy

Threads are grouped into blocks of constant size/shape.
Blocks are launched as a grid.
We choose the dimensions of the grid to control how many blocks are launched, and thus how many threads are launched.

Let's design a SGEMM kernel!

Recall

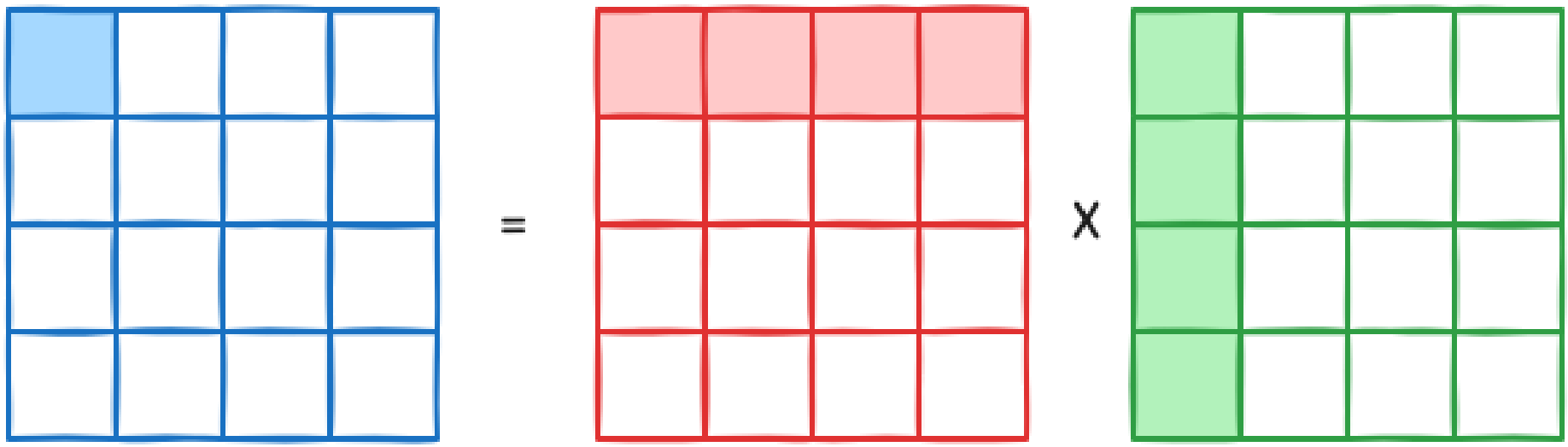
Recall that matrix multiplication is naturally parallel, and the output elements of C can be computed independently in parallel.



Let's design a SGEMM kernel!

Recall

Recall that matrix multiplication is naturally parallel, and the output elements of C can be computed independently in parallel.

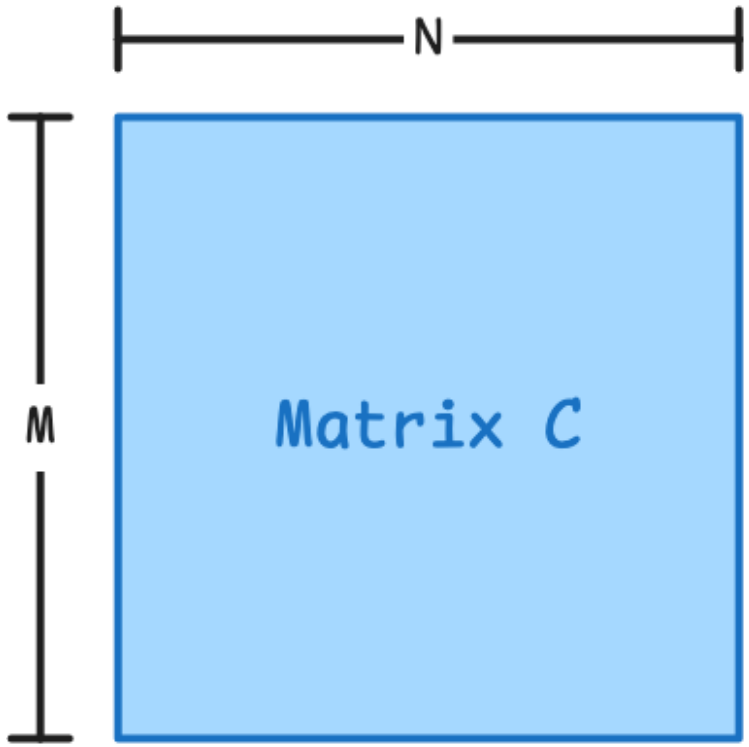


We can naturally map one thread to each output element!

Naive SGEMM Kernel

```
__global__ void sgemm_naive(
    int M, int N, int K, float alpha, const float *A,
    const float *B, float beta, float *C
) {
  const uint x = blockIdx.x * blockDim.x + threadIdx.x;
  const uint y = blockIdx.y * blockDim.y + threadIdx.y;

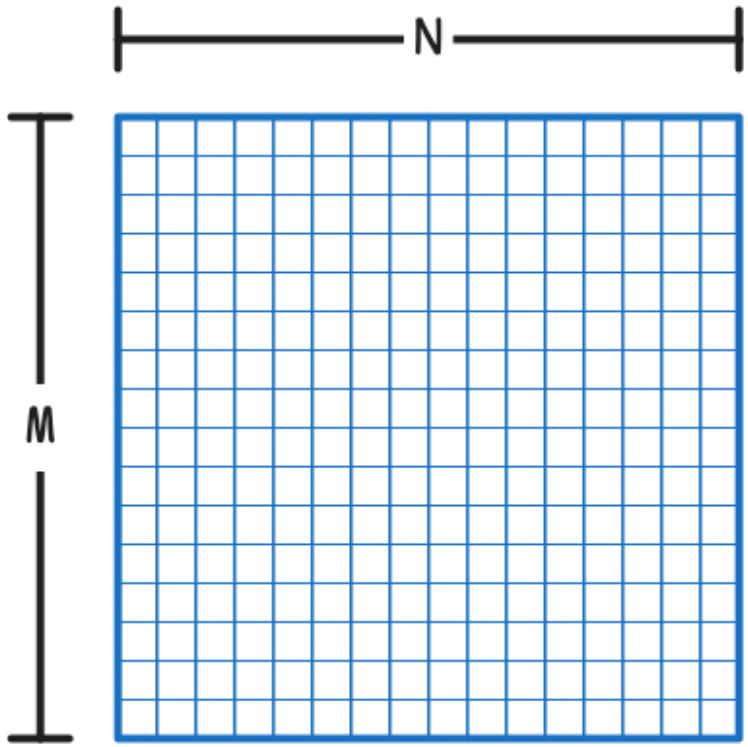
  if (x < M && y < N) {
    float tmp = 0.0;
    for (int i = 0; i < K; ++i)
      tmp += A[x*K+i] * B[i*N+y];
    C[x*N+y] = alpha*tmp + beta*C[x*N+y];
  }
}
void run_sgemm_naive(
  int M, int N, int K, float alpha, float *A,
  float *B, float beta, float *C
) {
  const int BM = 32, BN = 32;
  dim3 blockDim(BN, BM);
  dim3 gridDim(CEIL_DIV(M, BN), CEIL_DIV(N, BM));
  sgemm_naive<<<gridDim, blockDim>>>(
      M, N, K, alpha, A, B, beta, C);
}
```



Naive SGEMM Kernel

```
__global__ void sgemm_naive(
    int M, int N, int K, float alpha, const float *A,
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) {
  const uint x = blockIdx.x * blockDim.x + threadIdx.x;
  const uint y = blockIdx.y * blockDim.y + threadIdx.y;

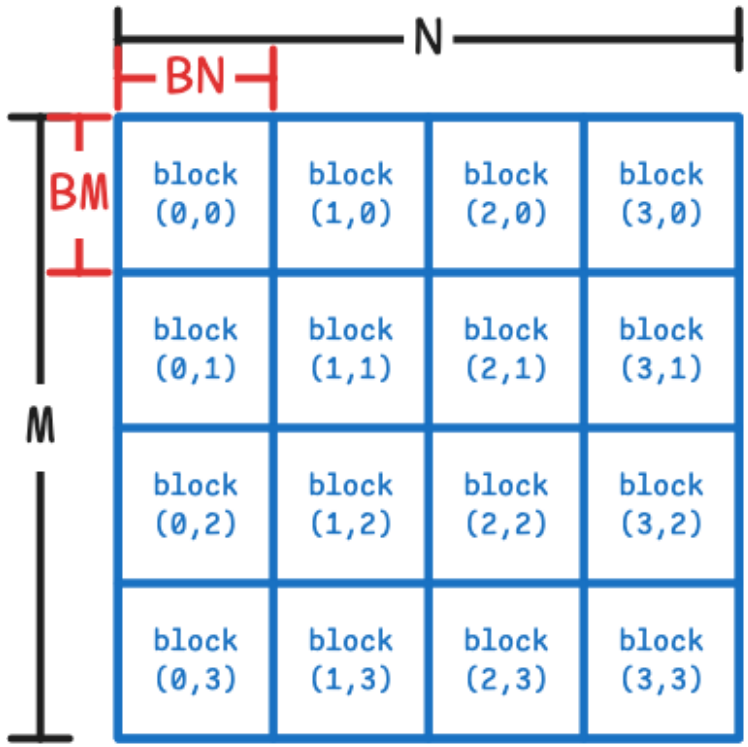
  if (x < M && y < N) {
    float tmp = 0.0;
    for (int i = 0; i < K; ++i)
      tmp += A[x*K+i] * B[i*N+y];
    C[x*N+y] = alpha*tmp + beta*C[x*N+y];
  }
}
void run_sgemm_naive(
  int M, int N, int K, float alpha, float *A,
  float *B, float beta, float *C
) {
  const int BM = 32, BN = 32;
  dim3 blockDim(BN, BM);
  dim3 gridDim(CEIL_DIV(M, BN), CEIL_DIV(N, BM));
  sgemm_naive<<<gridDim, blockDim>>>(
      M, N, K, alpha, A, B, beta, C);
}
```



Naive SGEMM Kernel

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__global__ void sgemm_naive(
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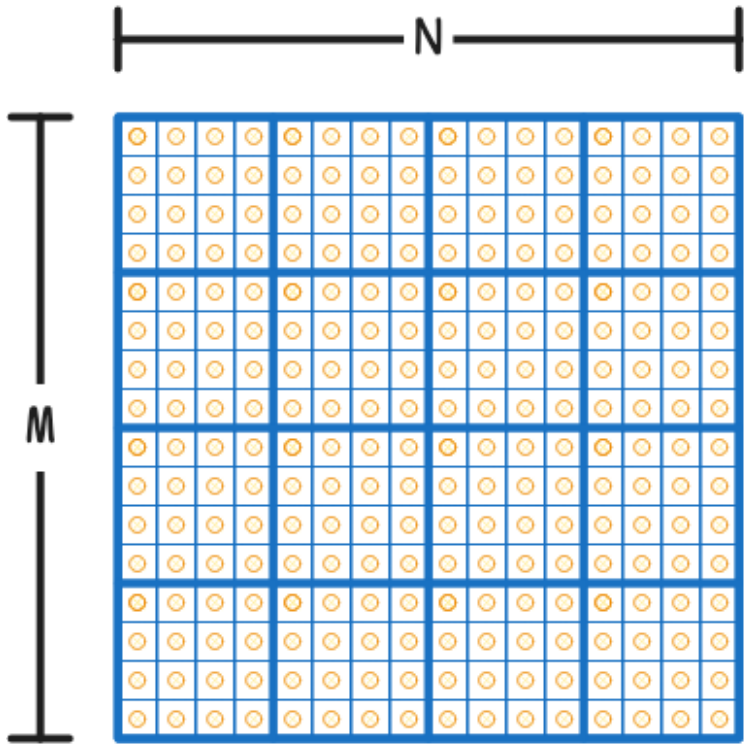
  if (x < M && y < N) {
    float tmp = 0.0;
    for (int i = 0; i < K; ++i)
      tmp += A[x*K+i] * B[i*N+y];
    C[x*N+y] = alpha*tmp + beta*C[x*N+y];
  }
}
void run_sgemm_naive(
  int M, int N, int K, float alpha, float *A,
  float *B, float beta, float *C
) {
  const int BM = 32, BN = 32;
  dim3 blockDim(BN, BM);
  dim3 gridDim(CEIL_DIV(M, BN), CEIL_DIV(N, BM));
  sgemm_naive<<<gridDim, blockDim>>>(
      M, N, K, alpha, A, B, beta, C);
}
```



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  const uint x = blockIdx.x * blockDim.x + threadIdx.x;
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  if (x < M && y < N) {
    float tmp = 0.0;
    for (int i = 0; i < K; ++i)
      tmp += A[x*K+i] * B[i*N+y];
    C[x*N+y] = alpha*tmp + beta*C[x*N+y];
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  const int BM = 32, BN = 32;
  dim3 blockDim(BN, BM);
  dim3 gridDim(CEIL_DIV(M, BN), CEIL_DIV(N, BM));
  sgemm_naive<<<gridDim, blockDim>>>(
      M, N, K, alpha, A, B, beta, C);
}
```



Naive SGEMM visualization

- One thread computes one element of C.
- It walks one row of A and one column of B for the full dot product.

Play naive-kernel visualization

Naive CUDA is already faster than the CPU

